A Systematic Literature Review of AI-Based Tools for SLRs and Meta-Analyses: A Comprehensive Overview

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Background: Systematic literature reviews (SLRs) and meta-analyses are critical methods for synthesizing research findings, but they are highly labor-intensive and time-consuming. AI technologies, particularly machine learning and deep learning, present an opportunity to automate significant portions of these processes, potentially accelerating the review timeline and reducing human error.

Objective: This study aims to provide a comprehensive overview of AI-based tools and frameworks that support the PRISMA (2020) SLR and meta-analysis processes.

Methods: A total of 128 unique AI-based tools were identified that support different phases of systematic reviews and meta-analyses. The tools were categorized according to the PRISMA (2020) framework and available performance metrics were extracted to assess their effectiveness in automating various steps of the review process.

Results: AI-based tools were found to be most prevalent in the Screening (79.7%) and Identification (53.1%) phases, with fewer tools addressing Data Extraction (37.5%), Data Synthesis (17.2%), Quality Assessment (5.5%), and Bias Assessment (7.0%). Notably 47.7% of tools supported multiple phases with most common overlaps in the combinations of Identification + Screening (40.6%) and Screening + Data Extraction (24.2%). Unsurprisingly Quality Assessment tools showed mostly overlaps with Data Extraction (71.4%) and Bias Assessment (57.1%).

Conclusion: The findings suggest a growing trend in AI tool development focused on the early phases of the systematic review process, especially Identification and Screening. Moreover, a broad spectrum of performance metrics have been used to assess the performance of said AI-based tools. However, more work is needed to evaluate these tools across the board as there is little to no consistency on how these tools are tested (different databases, versions, tools etc.). Lastly more work is needed for the later stages of review, such as Data Synthesis and Quality Assessment. Future research should focus on integrating AI solutions across the entire systematic review process and create a standardized testing methodology.

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Additional Key Words and Phrases: AI-based tools, tools, systematic reviews, meta-analysis, automation, artificial intelligence, prisma, machine learning

1 INTRODUCTION

As Rowley et al. [82] puts it eloquently "Scientific knowledge is based on the accumulated findings from prior research in a research domain, where individual studies constitute building blocks". In many domains it is commonly understood that new studies will build on previous research [31].However, there has been a significant rise in both education researchers and actual academic publications. From 1980 to 2018 the amount of education researches increased from 4 million to 13.1 million worldwide (3.3-fold increase) [93]. Meanwhile the number of academic publications increased from 0.65 million in 1980 to 3.16 million in 2018 (4.9-fold increase) [93]. This presents a challenge to researches meaning to build upon the existing knowledge in their research domains. It is simply not feasible for researches to have to go through the scientific body of their research domain each time.

Fortunately, this is not necessary as both systematic literature reviews and meta-analysis exist. Systematic literature reviews (SLRs) provide researches with an overview of a research domain in a single paper [71]. Similarly meta-analyses offer the opportunity to critically evaluate and statistically combine results of comparable studies or trials[23]. The latter plays a crucial role in evidence-based research by synthesizing findings from numerous studies and establishing statistical significance with studies that have conflicting results.

However both SLRs and Meta-analysis are very labor intensive and take long times to complete. Based on the PRISMA framework, which is widely endorsed, they require research questions, search strategy, study selection, data extraction and the synthesis of results [55]. The latest updated guideline of PRISMA 2020 framework comprises of a 27-item checklist [70]. As one can imagine the completion of said tasks takes a long time. Rohit et al. conducted an analysis on the estimated time

needed to complete 195 registered and completed reviews and found the mean time to be 67.3 weeks and the longest taking as long as 186 weeks [11]. There are several factors as to why SLRs and meta-analyses are inherently time-consuming. Firstly there are many manual processes involved, screening of studies, selection of studies and the data extraction. As these research studies encompass large parts or entire study domains, the amount of studies are usually in the thousands. Secondly, but perhaps most importantly this entire process must be reproducible by other researches to maintain scientific rigor and ensure quality [87]. Moreover researches have to include all relevant studies, regardless of quality and minimize bias [87]. Ensuring quality in that scenario, extends the duration of the review processes.

Artificial Intelligence is an research domain that continuous to be a subject of extensive debate and research. Especially surrounding the question "What is AI?". Throughout the years many different researches have attempted to define the term *artificial intelligence*. John McCarthy, seen by some as the father of modern Artificial Intelligence research was the first to attempt and define the term as "The science and engineering of making intelligent machines" [21]. Some decades later Elain Rich defined AI as "The study of how to make computer do things that people are better at" [80]. This expanded the definition to include activities such as natural language understanding, problem solving and doing science [80]. While there is no widely accepted definition of AI, it is generally agreed upon that AI refers to a field of computer science dedicated to having systems perform task that usually require human intelligence. Essentially the integration of human-like intelligence into machines [45].

As the definition expanded to include broader range of activities, distinct subsets of Artificial Intelligence emerged, (1) Machine Learning and (2) Deep Learning. Machine Learning is a subset of AI and Deep Learning is a subset of ML. Machine learning encompasses the methods that machine(s) learn from using data. Using provided data - usually large data - a machine learns to make decisions without explicit programming [45].

Deep Learning expands on traditional Machine Learning by emulating the way our brain processes information. It accomplishes this through artificial neural networks [49]. It is therefore sometimes referred to as *deep structured learning* or *hierarchical training* [4].

The continuous research and development of AI and it's subsets has opened up opportunities to automate several manual processes of SLRs and Meta-Analyses. There is potential for researches to save significant time and lower the entry barrier for new research. However despite the potential, this is still a relatively new area of application and a comprehensive overview of the current research is lacking. As a result, it is challenging to assess the current state of research and its future prospects.

This paper will analyze existing literature and conduct a meta-analysis to create a comprehensive overview of the existing tools for SLRs and Meta-Analyses. The aim is to answer the following research question: *What AI-based tools and frameworks exist to aid in the creation of Systematic Literature Reviews and/or Meta-Analyses?*

2 METHODS

2.1 Information Sources

The studies on which this meta analyis was performed were acquired using 2 different methods. Firstly certain tools/papers were includes based on expert guidance. Secondly a Scopus search was performed on 15th February 2025.

2.1.1 Expert. Based on several meetings and e-mail exchanges several tools/papers were manually selected to be included for the analysis.

2.1.2 Scopus. The full search query can be seen in the appendix. As the full search query is quite long for readability purposes a simplified version of the search query can be seen below:

```
(TITLE-ABS-KEY("systematic review" OR "meta-analysis"
     OR "evidence synthesis"))
AND (TITLE-ABS-KEY("machine learning" OR "artificial
     intelligence" OR "LLM"
     OR "ChatGPT" OR "natural language processing" OR "
          automation"))
AND (TITLE-ABS-KEY("tool*" OR "software" OR "framework
     *" OR "screening"
     OR "data extraction" OR "study selection"))
AND NOT (TITLE-ABS-KEY("clinical" OR "patient*" OR "RCT
     " OR "hardware" OR "robot*"
     OR "student*" OR "pedagog*" OR "laboratory" OR "
          experiment*"
     OR "covid*" OR "cancer*" OR "infection*"))
```

This search query resulted in 6464 records. These search terms were chosen to capture papers discussing tools that assist in one or more processes involved in creating SLRs and Meta-Analyses.

2.2 Eligibility criteria

Based on the inclusion and exclusion criteria, studies will be accepted or rejected. The criteria are as follows:

- Study must evaluate or describe a tool
- Tool must be accessible to researchers
- Tool has to be for meta-analysis and/or systematic reviews
- Tool has to assist in at least one of the steps of the PRISMA 2020 framework
- Tool should include some form of AI (Artificial Intelligence) or ML (Machine Learning) methodologies. Based on the definitions provided

- Manuscript for tool must be written in English, Dutch or Bosnian (Serbo-Croatian); the author is proficient in these 3 languages.

2.3 Search Strategy

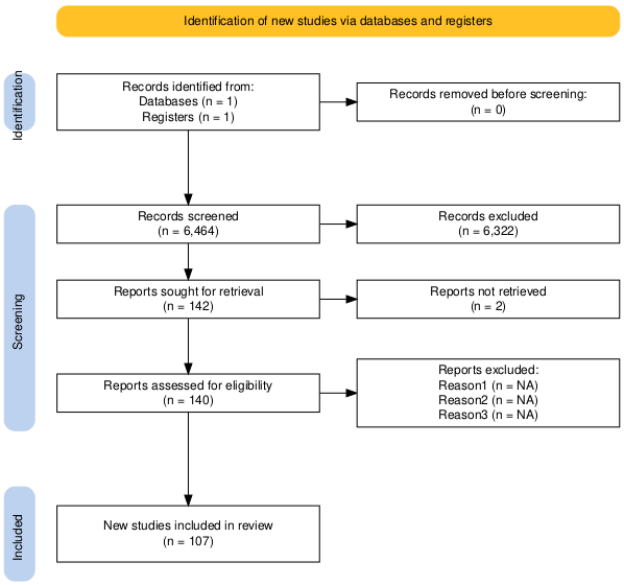

Fig. 1. Flow chart (Study Selection) [33]

2.4 Data Items

From the included papers, data variables were extracted as completely as possible.

2.4.1 . Codebook Variables

(1) Study Identifiers
  (a) **ID**; Identifier of the study (e.g., 1, 2, 3, . . .)
  (b) **Study**; Author name and year (e.g., Desolda22)
  (c) **Year**; Year of publication (e.g., 2017; 2003)
  (d) **Primary Reference**; Full reference (e.g., Smith, J., 2022. AI Tools. . .)
(2) Tool Identifiers
  (a) **Type**; Tool or framework (e.g., tool)
  (b) **Name**; Name of tool/framework (e.g., AsScreener)
  (c) **PRISMA 2020 Support**; Phase(s) of the PRISMA 2020 framework that the tool supports (e.g., Screening, Data Extraction)
  (d) **AI or Subset of AI**; Specifies if the tool is AI or a subset (e.g., AI, ML, DP, NLP, LLM)
(3) Performance and Functionality Identifiers
  (a) **Performance Metrics**; Performance metrics of the tool (e.g., Precision: 85%)
  (b) **Automation Level**; Level of automation (e.g., Semi-Automated)
  (c) **Standalone or Integrated**; Whether the tool is standalone or integrated (e.g., Standalone)
(4) Technical Identifiers
  (a) **Input Type**; Data input type for the tool (e.g., XML)
  (b) **Open or Closed Source**; Whether the tool/framework is open or closed source (e.g., Open Source)
  (c) **Import/Export Functionality**; Import/export functionality (e.g., CSV)
  (d) **Bibliography Compatibility**; Compatibility with bibliography formats (e.g., Yes)
  (e) **Domain Application**; Application domain (e.g., Healthcare)
  (f) **Output Produced**; Format or type of output produced (e.g., CSV)
(5) Effectiveness
  (a) **Effectiveness**; Summary of the tool's effectiveness (e.g., Reduces time by x%)

3 RESULTS

For readability purposes, a condensed and alphabetically ordered version of the extracted data is presented in the table below. This table provides an overview of the most relevant information and given that some tools were discussed across multiple studies, duplicate entries have been merged, with the primary reference listing all relevant studies. For those interested the full table can be seen in the Appendix.

PRISMA Phase Abbreviations and Performance Metrics

Abbrev.	PRISMA Phase	Performance Metrics (with Definitions)
I	Identification	–
S	Screening	• Sensitivity/Recall: Proportion of relevant documents correctly identified • Specificity: Proportion of irrelevant documents correctly excluded • Precision: Proportion of selected documents that are relevant • F1-score: Harmonic mean of precision and recall • WSS@R: Work Saved over Sampling at recall level R • AUC: Area Under the ROC Curve
DE	Data Extraction	• Accuracy: Proportion of correctly extracted data points • Completeness: Proportion of required data successfully extracted
DS	Data Synthesis	• Coherence Score: Measure of thematic consistency in synthesis • Similarity Score: Measure of conceptual alignment between studies
QA	Quality Assessment	• Agreement: Inter-rater reliability in quality judgments • Accuracy: Compared to expert human assessments
B	Bias Assessment	• Agreement: Consistency with established bias evaluation frameworks • Detection Rate: Proportion of actual biases correctly identified

Table 1. Condensed Systematic Review Tools Overview (128 Tools) [19]

Tool Name	References	Type	I	S	DE	DS	QA	B	AI Type	Performance Metrics
2weekSR	[81, 86]	tool	✓	✓	✓				ML	-
Abstrackr	[10, 27–29, 58, 67, 75, 81, 84, 100]	tool		✓					ML	sensitivity[28, 29], specificity[28], precision[27, 28]
Active Pubmed Search (APS)	[84]	tool	✓						ML	-
ASReview	[9, 10, 29, 75, 76, 81]	tool		✓					ML	-
Atlas Ti	[67]	tool			✓				-	-
Automatic NMA Screener	[53]	tool		✓					ML: NLP	sensitivity[53], workload decrease[53]
AutoLit	[38]	tool			✓				ML	-
BiblioShiny	[69]	tool	✓						ML: Text Mining	-
Carrot2	[69]	tool	✓						ML: Text Mining	-
ChatGPT (no version)	[32, 42, 46, 57, 104]	tool	✓	✓	✓				LLM	accuracy[42, 46], sensitivity[42]
ChatGPT3	[90]	framework		✓					LLM	sensitivity score[90]
ChatGPT3.5	[3, 20, 41, 54, 61, 64, 77, 83, 89]	tool	✓	✓					LLM	accuracy[54, 64, 83], specificity[54]
ChatGPT3.5-Turbo	[43]	tool		✓					LLM	precision[43], NPV[43]
ChatGPT4	[20, 32, 36, 52, 54, 57, 78, 92, 96]	tool	✓	✓	✓			✓	LLM	accuracy[32, 54], sensitivity[52, 54]
Chatpdf	[88]	tool			✓				ML	-
Citation Screener	[6, 50]	tool		✓					ML	precision[6, 50], recall[6]
Citation Screener (BioMedical)	[102]	tool		✓					ML	reduction manual screening[102]
CitNetExplorer	[69]	tool	✓						ML: Text Mining	-
Claude3	[54, 57, 78]	tool	✓	✓	✓				LLM	accuracy[54, 78]
Cochrane	[58, 84]	tool		✓					ML	-
Colandr	[10, 29, 81, 84]	tool	✓	✓					ML	-
Covidence	[29, 57, 67]	tool		✓					ML	-
Cruise Screening	[51]	tool		✓					ML	-
Data Extracter	[30]	tool			✓				ML	recall[30], precision[30]
Deducklick	[81]	tool	✓						ML	-

Continued on next page

Table 1 – continued

Tool Name	References	Type	I	S	DE	DS	QA	B	AI Type	Performance Metrics
Dextr	[10, 101]	tool			✓				ML	precision[101], recall[101]
DistillerSR	[10, 29, 57, 67, 81, 84]	tool	✓	✓	✓				ML	-
DocAnalyzer AI	[88]	tool		✓	✓				ML	-
DoCTER	[79]	tool		✓					ML	ensemble score[79]
Document Screener	[35]	tool		✓					ML	F1-score[35], accuracy[35]
Easy-peasy AI	[88]	tool							-	-
Elicit	[29, 44, 94, 95, 112]	tool	✓	✓	✓				LLM/ML	recall[112], precision[112]
EndNote	[69]	tool	✓						ML: Text Mining	-
EPPI-Reviewer	[10, 29, 57, 58, 67, 81, 84, 100]	tool	✓	✓		✓			ML	-
ExaCT	[10, 74, 81, 84]	tool			✓				ML: Text Mining	-
FAST2	[10]	tool		✓					ML	-
FastRead	[68]	tool							-	-
FSL-based screening	[108]	framework		✓					ML	workload reduction[108]
GAPScreener	[84]	tool		✓					ML	-
GATE	[81]	tool			✓				ML	-
Gemini 1.0	[54]	tool		✓					LLM	accuracy[54], specificity[54]
General-Purpose Filtering System	[47]	tool		✓					NLP	area under curve (AUC)[47]
Google PaLM2	[54]	tool		✓					LLM	accuracy[54], specificity[54]
GRADEpro	[67]	tool					✓		-	-
Hipdf	[88]	tool			✓				ML	-
IBM Parse	[84]	tool	✓						ML: Text Mining	-
In-Vitro Screener	[106]	tool		✓					ML: Text Mining	sensitivity[106], specificity[106]
Iris.ai	[10]	tool		✓	✓				ML	-
JBI-SUMARI	[67]	tool		✓	✓		✓		-	-
Keyword-Analyzer	[69]	tool	✓						ML: Text Mining	-
KeyCharacteristic Screener	[97]	tool		✓					ML: Text Mining	recall[97], exclusion[97]
Leximancer	[81]	tool	✓						ML	-
Literature Search Sandbox	[1]	tool	✓						LLM	sensitivity score[1]
LitSearchR	[81]	tool	✓						ML	-
LitSuggest	[10, 84]	tool	✓	✓					ML	-
LLama2	[54]	tool		✓					LLM	accuracy[54], specificity[54]
Llama3	[54]	tool		✓					LLM	accuracy[54], specificity[54]
LLM-FA	[16]	tool		✓					LLM	accuracy score[16]
LLM-SA	[16]	tool		✓					LLM	accuracy score[16]
Lingo3G	[69]	tool	✓						ML: Text Mining	-
Lingo4G	[69]	tool	✓						ML: Text Mining	-
MeSH on Demand	[67, 69]	tool	✓						ML: Text Mining	-
MetaCyto	[40]	tool			✓				ML	-
MetaMap	[81]	tool		✓					ML	-
MetaPreg	[84]	tool	✓	✓	✓				ML: NLP	-
ML-based screening	[17, 60, 72, 98]	tool	✓	✓					ML	specificity[60], ROC curves[60]
MMDas-AE	[84]	tool		✓					NLP	-
MySLR	[5]	tool	✓	✓	✓	✓			ML: Text Mining, NLP	-
NaCTeM	[58, 84]	tool			✓				ML: Text Mining	-
Natural Language Query's	[103]	tool	✓						NLP	recall[103], accuracy[103]
Nested Knowledge	[10, 29]	tool	✓	✓		✓			ML	-
NLP-based screening	[18, 63, 91, 110, 111]	tool	✓	✓	✓				NLP	recall[63], precision[63]
Nivvo	[67]	tool							-	-
PDFBox	[81]	tool			✓				ML	-
Paperfetcher	[81]	tool	✓						ML	-
Pdf2gpt	[88]	tool			✓				ML	-
PICOPortal	[10]	tool		✓					ML	-
Pitts.ai	[10]	tool		✓					ML	-

Continued on next page

Table 1 – continued

Tool Name	References	Type	I	S	DE	DS	QA	B	AI Type	Performance Metrics
Polygot	[67]	tool							-	-
PublicationBias	[81]	tool						✓	ML	-
PubReMiner	[67, 69]	tool	✓						ML: Text Mining	-
Pythia	[2]	tool		✓					ML: NLP	precision[2], sensitivity[2]
RAPTOR: RevMan	[58]	tool			✓				ML	-
RapidMiner	[81]	tool		✓					ML	-
RCT Tagger	[58, 81]	tool	✓						ML	-
Rayyan	[10, 29, 67, 75, 81, 84, 100]	tool		✓					ML	-
ResearchRabbit	[44]	tool	✓	✓					NLP	-
ResearchScreener	[10, 15, 67, 68, 84]	tool		✓					ML	workload saving[15]
RevTools	[67, 84]	tool				✓			NLP: Text Mining	-
RobuMeta	[81]	tool						✓	ML	-
RobotAnalyst	[10, 58, 67, 68, 81, 84]	tool		✓					ML	-
RobotReviewer	[10, 29, 58, 81, 84]	tool		✓	✓		✓	✓	ML	-
RobotSearcher	[58, 81]	tool	✓						ML	-
SciSpace	[29, 88, 94, 95, 112]	tool	✓	✓	✓				ML/LLM	recall[112], precision[112]
SciSpaceGPT	[29]	tool			✓		✓		DL	-
Screen Smarter not Harder	[14]	tool		✓					ML	recall[14], sensitivity[14]
Screen4Me	[65, 84]	tool	✓	✓					ML	agreement[65], sensitivity[65]
SearchRefiner	[29]	tool	✓						ML	-
SLR Qub	[81]	tool	✓						ML	-
SLuRp	[12]	tool							-	-
SMAFIRA	[13]	tool	✓						DL	-
SPARK	[7]	tool	✓	✓	✓				NLP (LDA)	coherence score[7]
srBERT	[8, 37]	tool		✓					ML: NLP	recall[8, 37], F1-score[8]
SRToolBox	[84]	tool	✓	✓	✓	✓			ML	-
SWARM-SLR	[107]	framework	✓	✓	✓	✓	✓	✓	AI	-
SWIFT-Active Screener	[10, 29, 56, 67, 84]	tool		✓					ML	agreement[56]
SWIFT-Review	[10, 29, 39, 58, 81, 84]	tool	✓	✓					ML	WSS@R[39]
SYMBALS	[65]	tool		✓					ML: Active Learning	speed (factor 6)[65]
SyRF	[81]	tool		✓					ML	-
SyROCCo	[24]	tool		✓	✓	✓			ML	similarity score[24]
SysRev	[10]	tool		✓					ML	-
Systematic Review Accelerator	[67]	tool		✓	✓				-	-
TerMine	[69, 81]	tool	✓						ML: Text Mining	-
Text Analyzer	[69]	tool	✓						ML: Text Mining	-
Text Classifier	[26]	tool		✓					ML	precision[26], recall[26]
Thalia	[58]	tool	✓						ML	-
TIAB-Screener	[29, 73]	tool		✓					ML	area under curve[73]
Tm for R	[69]	tool	✓						ML: Text Mining	-
Transformer	[25]	tool		✓					ML	workload saved (WSS@R)[25]
Trial2Rev	[59]	tool	✓	✓	✓				ML	-
Twister	[48, 84]	tool	✓	✓					ML: Text Mining	-
VosViewer	[69]	tool	✓						ML: Text Mining	-
Voyant	[69]	tool	✓						ML: Text Mining	-
WordNet	[81]	tool	✓						ML	-
Yale MESH	[67, 69]	tool	✓						ML: Text Mining	-
Zettair	[81]	tool	✓						ML	-

We identified a total of **128 unique tools** [1–3, 5–10, 12–18, 20, 24–30, 32, 35–44, 46–48, 50–54, 56–61, 63–65, 67–69, 72, 74–79, 81, 83, 84, 86, 88–92, 94–98, 100–104, 106–108, 110–112].

- **Identification (I)**: 68 tools (53.1%)
- **Screening (S)**: 102 tools (79.7%)
- **Data Extraction (DE)**: 48 tools (37.5%)
- **Data Synthesis (DS)**: 22 tools (17.2%)
- **Quality Assessment (QA)**: 7 tools (5.5%)
- **Bias Assessment (B)**: 9 tools (7.0%)

4 DISCUSSION

128 unique AI-based tools were identified all designed to support one or multiple phases of the SLR/Meta-analysis process. The distribution of the tools across the PRISMA 2020 phases reveal a strong emphasis on screening (79.7%), followed by identification (53.1%) and data extraction (37.5%), while data synthesis (17.2%), quality assessment (5.5%) and bias assessment (7.0%) are underrepresented. A critical insight found is that nearly half (47.7%, n=61) of tools supported multiple phases, but with striking imbalances. The most common combinations were Identification & Screening (40.6%, n=52), Screening & Data Extraction (24.2%, n=31) and Identification & Screening & Data Extraction (14.1%, n=18). This likely has to do with the fact that most of these phases share technical techniques, for example NLP (Natural Language Processing) for text processing. While the overlap of phases such as screening and data extraction suggests researches/developers are beginning to bridge distinct phases and possibly create pipelines, there is still a clear lack of tools covering the latter part of the SLR/Meta-Analysis process.

Only a singular framework was identified that covered the entire process (SWARM-SLR) and the only tool covering multiple initial and later phases was RobotReviewer. Unsurprisingly this suggests that most tools prioritize speeding up the phase and focus less on the methodological rigor. This makes sense as the most time consuming phases are the most manually labor intensive ones, these include the same repetitive task over and over again and are easier to automate using artificial intelligence due to the lack of variables. Meanwhile the latter phases require a lot of human touch and may therefore be significantly harder to automate, leading to lack of tools.

To ensure that these tools perform adequately enough that they pass the quality requirements, they need to be tested. Numerous studies introduced performance metrics, in particular tools in the earlier phases like those that covered identification, screening and data extraction. However it is hard to compare even a tool covering the same phase using the same performance metrics as tools may use different AI-techniques, use different datasets or different versions of the same LLM. It is therefore necessary to create a standardized pipeline for testing tools across numerous phases of the SLR on numerous domains. For future work on top of an standardized benchmark (tool/framework), an open-source unified architecture that covers all phases should be introduced. Finally the limitation of this study are as follows. While an attempt was made to make the search query as broadly as possible, it is still possible that the search may have missed relevant papers. Secondly the screening and data extraction was done by a single person and human's are prone to error and therefore the author could have missed relevant papers during the screening process or data extraction phase. Thirdly the review relied on the reported metrics of the papers and conducted no hands on testing, which may have led to overestimating or underestimating certain tools and many tools did not report any key metrics leading to a gap in performance data. Validation is necessary and should be the next step.

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Appendix: Full Search Query & Full Data Extraction Table

(TITLE-ABS-KEY ("systematic review") OR TITLE-ABS-KEY ("meta-analysis") OR TITLE-ABS-KEY ("meta analysis") OR TITLE-ABS-KEY ("systematic literature review") OR TITLE-ABS-KEY ("evidence synthesis")) AND (TITLE-ABS-KEY ("artificial intelligence") OR TITLE-ABS-KEY ("AI") OR TITLE-ABS-KEY ("machine learning") OR TITLE-ABS-KEY ("large language model*") OR TITLE-ABS-KEY ("LLM") OR TITLE-ABS-KEY ("natural language processing") OR TITLE-ABS-KEY ("ChatGPT") OR TITLE-ABS-KEY ("automation")) AND (TITLE-ABS-KEY (tool*) OR TITLE-ABS-KEY ("software") OR TITLE-ABS-KEY ("framework*") OR TITLE-ABS-KEY ("screening") OR TITLE-ABS-KEY ("data extraction") OR TITLE-ABS-KEY ("study selection")) AND NOT (TITLE-ABS-KEY ("randomized controlled trial") OR TITLE-ABS-KEY ("RCT") OR TITLE-ABS-KEY ("patient*") OR TITLE-ABS-KEY ("clinical") OR TITLE-ABS-KEY ("diagnos*") OR TITLE-ABS-KEY ("therap*") OR TITLE-ABS-KEY ("surg*") OR TITLE-ABS-KEY ("pharmaco*")
OR TITLE-ABS-KEY ("hardware") OR TITLE-ABS-KEY ("circuit*") OR TITLE-ABS-KEY ("sensor*") OR TITLE-ABS-KEY ("robot*") OR TITLE-ABS-KEY ("IoT")
OR TITLE-ABS-KEY ("curriculum") OR TITLE-ABS-KEY ("pedagog*") OR TITLE-ABS-KEY ("student*")
OR TITLE-ABS-KEY ("statistical method*") OR TITLE-ABS-KEY ("laboratory") OR TITLE-ABS-KEY ("experiment*")) AND NOT (TITLE-ABS-KEY ("covid*") OR TITLE-ABS-KEY ("corona") OR TITLE-ABS-KEY ("disease*") OR TITLE-ABS-KEY ("tumor*") OR TITLE-ABS-KEY ("cancer*") OR TITLE-ABS-KEY ("infection*"))

Table 2. Complete Data Extraction of All Systematic Review Tools (128 Tools)

ID	Study	Year	Primary Ref	Tool Name	Type	PRISMA Support	AI Type	Performance Metrics	Automation	Integration	Input Type	Source	Import/Export	Bibliocompat	Domain	Output	Effectiveness
31	David24	2024	[18]	NLP-based screening	framework	Screening, Data Extraction	NLP	Sentiment accuracy	Semi-automated	Standalone	Study data set	Open	No	No	Software engineering		Accuracy rate (86.5%)
83	Atkinson24	2024	[7]	SPARK	tool	Identification, Screening, Data Extraction	NLP (LDA)	Coherence score	Semi-automated	Standalone	Text	Open	Yes	No	Not specified	CSV	
138	Marshall24	2024	[90]	GPT-3 Prompt Templates	framework	Screening	LLM	Sensitivity score	Semi-automated	Standalone	Text	Open	No	No	Systematic reviews of RCTs	Text	Screening sensitivity (97.7%)
162	Ashleigh24	2024	[79]	DoCTER	tool	Screening	ML	Ensemble score	Semi-automated	Standalone	Text		No	No	Health	Text	
194	Chen24	2024	[16]	LLM-FA	tool	Screening	LLM	Accuracy score	Automated	Standalone	Text (prompts)	Open	No	No	Cross-disciplinary	Text	22.7%, 19.4%, 6.1%
194	Chen24	2024	[16]	LLM-SA	tool	Screening	LLM	Accuracy score	Semi-automated	Standalone	Text (papers)	Open	No	No	Cross-disciplinary	Text	22.7% (included), 22.2% (excluded)
206	Jensen24	2024	[62]	ChatGPT4-based screening	tool	Data Extraction	LLM	Accuracy score	Automated	Standalone	Text	Open	No	No	Not specified	Text	22.4% accuracy (5.2% false data)
320	Waltzod24	2024	[105]	Search Strategy	framework	Identification											
383	Dukic24	2024	[112]	Elicit	tool	Identification, Screening	LLM	Recall, precision	Automated	Integrated	Text	Closed	Yes	Yes	Software engineering	Text	18.3% precision
383	Dukic24	2024	[112]	SciSpace	tool	Identification, Screening	LLM	Recall, precision	Automated	Integrated	Text, PDF	Closed	Yes	Yes	Software engineering	Text	18.8% precision
392	Jiang24	2024	[46]	ChatGPT-based screening	tool	Identification, Screening, Data Extraction	LLM	Accuracy score	Semi-automated	Standalone	Text	Open	No	No	Agriculture	Text document	0.941
418	Lixia24	2024	[29]	Abstrackr	tool	Screening	ML										
418	Lixia24	2024	[29]	DistillerSR	tool	Identification, Screening, Data Extraction	ML				Title, abstract	Closed					
418	Lixia24	2024	[29]	RobotReviewer	tool	Data Extraction, Quality Assessment	NLP				Title, abstract	Closed					
418	Lixia24	2024	[29]	Rayyan	tool	Screening	ML				Abstract, title, keywords	Closed					
418	Lixia24	2024	[29]	EPPI-Reviewer	tool	Screening, Data Synthesis	ML				Abstract, title, keywords	Closed					
418	Lixia24	2024	[29]	ASReview	tool	Screening	ML				Text	Open					
418	Lixia24	2024	[29]	Covidence	tool	Screening	ML					Closed					
418	Lixia24	2024	[29]	SWIFT-Review	tool	Identification, Screening, Data Extraction, Data Synthesis	ML					Open					
418	Lixia24	2024	[29]	SearchRefiner	tool	Identification	ML				References	Open					
418	Lixia24	2024	[29]	Elicit	tool	Identification, Data Extraction, Data Synthesis	NLP				Title, abstract, PDFs	Closed					
418	Lixia24	2024	[29]	Nested Knowledge	tool	Identification, Screening, Data Synthesis	ML				Keywords	Closed					
418	Lixia24	2024	[29]	SWIFT-Active Screener	tool	Screening	ML				Abstract, title, keywords	Closed					
418	Lixia24	2024	[29]	Colandr	tool	Screening, Data Extraction	ML				Abstract, title, keywords	Closed					
418	Lixia24	2024	[29]	SciSpace GPT	tool	Data Extraction, Quality Assessment	DL				Papers	Closed					
418	Lixia24	2024	[29]	SciSpace	tool	Screening, Data Extraction	ML				Text	Closed					
451	Pilz24	2024	[73]	TIAB-Screener	tool	Screening	ML	Area under curve	Semi-automated	Standalone	Text	Open	No	No	Not specified		AUC table available

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Table 2 – continued

ID	Study	Year	Ref	Tool Name	Type	PRISMA Support	AI Type	Performance Metrics	Automation	Integration	Input Type	Source	Import/Export	Biblio	Domain	Output	Effectiveness
543	Issaiy24	2024	[42]	ChatGPT-based screening	framework	Screening	LLM	Sensitivity score, AUC, PPV & NPV, CIs	Automated	Standalone	Text	Closed	No	No	Radiology		Sensitivity 95%, NPV 99%
550	Teperikidis24	2024	[92]	ChatGPT4-based screening	framework	Identification, Screening	LLM		Semi-automated	Standalone	Text	Closed	No	No	Not specified		
584	Menold24	2024	[60]	ML-enabled screening	framework	Screening	ML	Specificity, ROC curves, missed studies, work saved	Semi-automated	Standalone	Studies	Closed	No	No	Not specified		
589	Li24	2024	[54]	ChatGPT v4-based screening	tool	Screening	LLM	Sensitivity, accuracy	Automated	Standalone	Abstract, title	Closed	No	No	Not specified		96.1% accuracy, 96% specificity
589	Li24	2024	[54]	ChatGPT v3.5-based screening	tool	Screening	LLM	Sensitivity, accuracy	Automated	Standalone	Abstract, title	Closed	No	No	Not specified		84.5% accuracy, 87% specificity
589	Li24	2024	[54]	Google PaLM 2-based screening	tool	Screening	LLM	Sensitivity, accuracy	Automated	Standalone	Abstract, title	Closed	No	No	Not specified		90% accuracy, 95% specificity
589	Li24	2024	[54]	Llama2-based screening	tool	Screening	LLM	Sensitivity, accuracy	Automated	Standalone	Abstract, title	Closed	No	No	Not specified		68% accuracy, 61% specificity
589	Li24	2024	[54]	Gemini 1.0-based screening	tool	Screening	LLM	Sensitivity, accuracy	Automated	Standalone	Abstract, title	Closed	No	No	Not specified		97% accuracy, 99% specificity
589	Li24	2024	[54]	Llama3-based screening	tool	Screening	LLM	Sensitivity, accuracy	Automated	Standalone	Abstract, title	Closed	No	No	Not specified		91% accuracy, 89% specificity
589	Li24	2024	[54]	Claude3-based screening	tool	Screening	LLM	Sensitivity, accuracy	Automated	Standalone	Abstract, title	Closed	No	No	Not specified		92% accuracy, 94% specificity
629	Boateng24	2024	[68]	FastRead	tool												
629	Boateng24	2024	[68]	Research Screener	tool		ML		Semi-automated	Standalone	Titles	Open					
629	Boateng24	2024	[68]	Robot Analyst	tool												
634	Thomas24	2024	[64]	ChatGPT v3.5-based screening	tool	Screening	LLM	Accuracy score	Semi-automated	Standalone	Abstract, title	Closed	No	No	Ecological		83% accuracy
634	Liu24	2024	[56]	SWIFT-Active Screener	tool	Screening	ML	Agreement	Automated	Standalone	Abstract, title	Closed	No	No			97.9% agreement
721	Han24	2024	[34]	RAG-Based screening	framework	Identification, Screening, Data Extraction, Data Synthesis	LLM		Semi-automated	Standalone	Text	Closed	No	No	Not specified		
750	Adam24	2024	[1]	Literature search sandbox	tool	Identification	LLM	Sensitivity score	Semi-automated	Standalone	Text	Open	No	No	Not specified		95% median sensitivity
766	Fang24	2024	[24]	SyROCCo	tool	Screening, Data Extraction, Data Synthesis	ML	Similarity score	Automated	Standalone	Text	Closed	No	No	Not specified		0.711
863	Bolanos24	2024	[10]	Abstrackr	tool	Screening	ML										
863	Bolanos24	2024	[10]	Colandr	tool	Screening	ML										
863	Bolanos24	2024	[10]	Covidence	tool	Screening	ML										
863	Bolanos24	2024	[10]	EPPI-Reviewer	tool	Screening	ML										
863	Bolanos24	2024	[10]	Rayyan	tool	Screening	ML										
863	Bolanos24	2024	[10]	SWIFT-Active Screener	tool	Screening	ML										
863	Bolanos24	2024	[10]	DistillerSR	tool	Screening	ML										
863	Bolanos24	2024	[10]	SWIFT-REVIEW	tool	Screening	ML										
863	Bolanos24	2024	[10]	SysRev	tool	Screening	ML										
863	Bolanos24	2024	[10]	Nested Knowledge	tool	Screening	ML										
863	Bolanos24	2024	[10]	RobotAnalyst	tool	Screening	ML										
863	Bolanos24	2024	[10]	LitSuggest	tool	Screening	ML										
863	Bolanos24	2024	[10]	Pitts.ai	tool	Screening	ML										
863	Bolanos24	2024	[10]	Research Screener	tool	Screening	ML										
863	Bolanos24	2024	[10]	ASReview	tool	Screening	ML										
863	Bolanos24	2024	[10]	PICOPortal	tool	Screening	ML										
863	Bolanos24	2024	[10]	FAST2	tool	Screening	ML										
863	Bolanos24	2024	[10]	Dextr	tool	Data Extraction	ML										
863	Bolanos24	2024	[10]	ExaCT	tool	Data Extraction	ML										
863	Bolanos24	2024	[10]	Iris.ai	tool	Screening, Data Extraction	ML										

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Table 2 – continued

ID	Study	Year	Ref	Tool Name	Type	PRISMA	Support	AI Type	Performance Metrics	Automation	Integration	Input Type	Source	Import/Export	Biblio	Domain	Output	Effectiveness
863	Bolanos24	2024	[10]	RobotReviewer/RobotSearch	tool	Screening, Extraction	Data	ML										
910	Landschaft24	2024	[52]	ChatGPT4-Based Screening	tool	Screening		LLM	Agreement, accuracy, sensitivity, specificity, F1-score	Automated	Standalone	Text	Closed	No	No	Healthcare		0.899, 0.975, 0.987, 0.950
1008	Butzke24	2024	[13]	SMAFIRA	tool	Identification		DL		Semi-automated	Standalone	Reference	Closed	No	No	Animal research		
1009	Spillias23	2023	[89]	ChatGPT-based screening	framework	Screening		LLM		Semi-automated	Standalone	Text	Closed	No	No			
1072	Syriani24	2024	[43]	ChatGPT-based screening	tool	Screening		LLM	Precision, NPV, specificity	Semi-automated	Standalone	Text	Closed	No	No			
1078	Ofori-Boateng24	2024	[67]	Covidence	tool	Screening, Extraction	Data			Semi-automated								
1078	Ofori-Boateng24	2024	[67]	JBI-SUMARI	tool	Screening, Extraction	Data			Semi-automated								
1078	Ofori-Boateng24	2024	[67]	Rayyan	tool	Screening												
1078	Ofori-Boateng24	2024	[67]	GRADEpro	tool													
1078	Ofori-Boateng24	2024	[67]	Nvivo	tool													
1078	Ofori-Boateng24	2024	[67]	Systematic Review Accelerator	tool	Screening, Extraction	Data			Semi-automated								
1078	Ofori-Boateng24	2024	[67]	Polygot	tool													
1078	Ofori-Boateng24	2024	[67]	Abstrackr	tool													
1078	Ofori-Boateng24	2024	[67]	DistillerSR	tool													
1078	Ofori-Boateng24	2024	[67]	Yale MESH Analyser	tool													
1078	Ofori-Boateng24	2024	[67]	MeSH on Demand	tool													
1078	Ofori-Boateng24	2024	[67]	EPPI-Reviewer	tool	Screening				Semi-automated								
1078	Ofori-Boateng24	2024	[67]	RobotAnalyst	tool	Screening				Semi-automated								
1078	Ofori-Boateng24	2024	[67]	RobotReviewer	tool													
1078	Ofori-Boateng24	2024	[67]	Atlas Ti	tool													
1078	Ofori-Boateng24	2024	[67]	PubReMiner	tool													
1078	Ofori-Boateng24	2024	[67]	Research Screener	tool	Screening				Semi-automated								
1078	Ofori-Boateng24	2024	[67]	RevTools	tool													
1078	Ofori-Boateng24	2024	[67]	SWIFT-ACTIVE Review	tool	Screening				Semi-automated								
1120	Chernikova24	2024	[17]	ML-enabled screening	tool	Screening		ML	Accuracy, recall, precision, F-score	Semi-automated	Standalone	Abstract, title	Closed	No	No			See table
1145	Hawkins24	2024	[47]	General-purpose filtering system	tool	Screening		NLP	Area under curve (AUC)	Semi-automated	Standalone	Abstract, title		No	No			See table
1163	Huotala24	2024	[41]	ChatGPT3.5-based screening	tool	Screening		LLM	Accuracy	Semi-automated	Standalone	Abstract, title		No	No			69
1239	Park24	2024	[110]	NLP-based screening	tool	Identification		NLP		Semi-automated	Standalone	Abstract, title		No	No			
1241	Trevino-Juarez24	2024	[96]	ChatGPT4-Based Screening	tool	Bias Assessment		LLM		Semi-automated	Standalone			No	No			
1334	Napoleao24	2024	[63]	ML-enabled screening	tool	Screening		ML	Recall, precision	Automated	Standalone	Abstract, title		No	No	System Engineering		14%, 15%

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Table 2 – continued

ID	Study	Year	Ref	Tool Name	Type	PRISMA Support	AI Type	Performance Metrics	Automation	Integration	Input Type	Source	Import/Export	Biblio	Domain	Output	Effectiveness
1337	Ye24	2024	[109]	Hybrid Workflow	framework	Identification, Screening, Data Extraction	ML	Accuracy	Semi-automated	Standalone	Abstract, title, full text						100% accuracy
1415	Demir24	2024	[20]	ChatGPT-based screening	framework	Identification	ML	Accuracy	Semi-automated	Standalone	Abstract, title		No	No			See table
1478	Quan24	2024	[76]	ASReview	tool	Screening	ML		Semi-automated		Dataset		Yes	No			
1499	Campos24	2024	[14]	Screening Smarter not Harder	tool	Screening	ML	Recall, sensitivity, specificity, time saving	Semi-automated	Standalone	Papers		No	No			
1508	Pijls24	2024	[72]	ML-enabled screening	tool	Identification	ML	Recall, confidence interval	Semi-automated	Standalone	Text		No	No	Orthopedics		
1725	Wiwatthanasettharn24	2024	[108]	FSL-based screening	framework	Screening	ML	Workload reduction, cosine similarity, recall needs	Automated		Abstract, title	Closed	No	No			31.11% workload reduction
1734	Bruggemann24	2024	[94]	Elicit	tool	Identification, Screening	ML										
1734	Bruggemann24	2024	[94]	SciSpace	tool	Identification, Screening	ML										
1860	Quan24	2024	[75]	Abstrackr	tool	Screening	ML				Abstract, title				Second language acquisition		
1860	Quan24	2024	[75]	ASReview	tool	Screening	ML				Abstract, title						
1860	Quan24	2024	[75]	Rayyan	tool	Screening	ML				Abstract, title						
1980	Wittenborg24	2024	[107]	SWARM-SLR	tool	Identification, Screening, Data Extraction, Data Synthesis, Bias	AI		Semi-automated	Integrated		Open	Yes	No	General	SLR	
2115	Tomczyk24	2024	[95]	Elicit	tool	Identification, Screening	ML										
2115	Tomczyk24	2024	[95]	SciSpace	tool	Identification, Screening	ML										
2248	Luo24	2024	[57]	EPPI-Reviewer	tool	Screening	ML										
2248	Luo24	2024	[57]	DistillerSR	tool	Screening	ML										
2248	Luo24	2024	[57]	ChatGPT-based screening	tool	Identification, Screening, Data Extraction	LLM										
2248	Luo24	2024	[57]	Claude3-based screening	tool	Identification, Screening, Data Extraction	LLM										
2258	Oami24	2024	[66]	ChatGPT4 Turbo-Based Screening	tool	Screening	LLM	Sensitivity, specificity	Automated	Standalone	Abstract, title	Closed	No	No	Japanese sepsis guidelines		Sensitivity 0.75, specificity 0.99
2304	Soufi24	2024	[88]	Chatpdf	tool	Data Extraction	ML		Automated	Integrated	PDF	Closed	Yes	No			
2304	Soufi24	2024	[88]	Pdf2gpt	tool	Data Extraction	ML		Automated	Integrated	PDF	Closed	Yes	No			
2304	Soufi24	2024	[88]	Hipdf	tool	Data Extraction	ML		Automated	Integrated	PDF	Closed	Yes	No			
2304	Soufi24	2024	[88]	SciSpace	tool	Screening, Data Extraction	ML		Semi-automated	Standalone	Text	Closed	No	No			
2304	Soufi24	2024	[88]	Easy-peasy AI	tool												
2304	Soufi24	2024	[88]	DocAnalyzer AI	tool	Screening, Data Extraction	ML		Semi-automated	Standalone	Text	Closed	No	No			
2343	Ruksakulpiwat24	2024	[83]	ChatGPT-based screening	tool	Identification, Screening	LLM	Precision	Semi-automated	Standalone	Text	Closed	No	No	Health		2.86
2422	Hasan24	2024	[36]	ChatGPT4-Based Screening	framework	Bias	LLM	Risk of bias	Semi-automated	Integrated	Text	Closed	No	No			
2463	Khan24	2024	[78]	ChatGPT4-Turbo-Based Screening	tool	Data Extraction	LLM	Accuracy	Automated	Standalone	Text	Closed	No	No	Clinical trials		
2463	Khan24	2024	[78]	Claude3-based screening	tool	Data Extraction	LLM	Accuracy	Automated	Standalone	Text	Closed	No	No	Clinical trials		
2583	Guo23	2023	[32]	ChatGPT-Based Screening	tool	Screening	LLM	Accuracy, sensitivity, F1-score	Automated	Standalone	Abstract, title	Closed	No	No	See table		See table
2583	Guo23	2023	[32]	ChatGPT4-Based Screening	tool	Screening	LLM	Accuracy, sensitivity, F1-score	Automated	Standalone	Abstract, title	Closed	No	No	See table		

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ID	Study	Year	Ref	Tool Name	Type	PRISMA Support	AI Type	Performance Metrics	Automation	Integration	Input Type	Source	Import/Export	Biblio	Domain	Output	Effectiveness
2709	Quershi23	2023	[77]	ChatGPT3.5-based screening	tool	Screening	LLM		Automated	Standalone	Abstract, title	Closed	No	No			
2715	Qang23	2023	[103]	Natural Language Query's	tool	Identification	NLP	Recall, accuracy	Semi-automated	Standalone	Text	Closed	No	No			See table
2737	Kusa23	2023	[51]	CRUISE-Screening	tool	Screening	ML		Automated	Integrated		Open	Yes	Yes			
2883	Roberts23	2023	[61]	ChatGPT3-based screening	tool	Identification	LLM	Mean difference	Automated	Standalone	Text	Closed	No	No			4.92% difference
2933	Roth23	2023	[81]	RCT tagger	tool	Identification	ML		Automated								
2933	Roth23	2023	[81]	Litsearchr	tool	Identification	ML		Automated								
2933	Roth23	2023	[81]	WordNet	tool	Identification	ML		Automated								
2933	Roth23	2023	[81]	RobotSearch	tool	Identification	ML		Automated								
2933	Roth23	2023	[81]	Zettair	tool	Identification	ML		Automated								
2933	Roth23	2023	[81]	SLR.qub	tool	Identification	ML		Automated								
2933	Roth23	2023	[81]	TerMine	tool	Identification	ML		Automated								
2933	Roth23	2023	[81]	Leximancer	tool	Identification	ML		Automated								
2933	Roth23	2023	[81]	Paperfetcher	tool	Identification	ML		Automated								
2933	Roth23	2023	[81]	Deducklick	tool	Identification	ML		Automated								
2933	Roth23	2023	[81]	DistillerSR	tool	Screening	ML		Automated								
2933	Roth23	2023	[81]	Rayyan	tool	Screening	ML		Automated								
2933	Roth23	2023	[81]	Colandr	tool	Screening	ML		Automated								
2933	Roth23	2023	[81]	Abstrackr	tool	Screening	ML		Automated								
2933	Roth23	2023	[81]	EPPI-Reviewer	tool	Screening	ML		Semi-automated								
2933	Roth23	2023	[81]	ASReview	tool	Screening	ML		Automated								
2933	Roth23	2023	[81]	RobotAnalyst	tool	Screening	ML		Semi-automated								
2933	Roth23	2023	[81]	SWIFT-Review	tool	Screening, Data Extraction	ML		Semi-automated								
2933	Roth23	2023	[81]	MetaMap	tool	Screening	ML		Automated								
2933	Roth23	2023	[81]	RapidMiner	tool	Screening	ML		Automated								
2933	Roth23	2023	[81]	SyRF	tool	Screening	ML		Automated								
2933	Roth23	2023	[81]	PDFBox	tool	Data Extraction	ML		Automated								
2933	Roth23	2023	[81]	GATE	tool	Data Extraction	ML		Automated								
2933	Roth23	2023	[81]	EXACT	tool	Data Extraction	ML		Automated								
2933	Roth23	2023	[81]	RobotReviewer	tool	Data Synthesis	ML		Automated								
2933	Roth23	2023	[81]	Robumeta	tool	Bias	ML		Automated	Integrated							
2933	Roth23	2023	[81]	PublicationBias	tool	Bias	ML		Automated	Integrated							
2933	Roth23	2023	[81]	2weekSR	tool	Identification, Screening, Data Extraction	ML		Semi-automated								
3037	Santos23	2023	[84]	Abstrackr	tool	Screening	ML		Semi-automated								
3037	Santos23	2023	[84]	Active Pubmed Search (APS)	tool	Identification	ML		Semi-automated								
3037	Santos23	2023	[84]	Cochrane RCT classifier	tool	Screening	ML		Automated						RCTs		
3037	Santos23	2023	[84]	Colandr	tool	Identification	ML: NLP		Semi-automated								
3037	Santos23	2023	[84]	DistillerSR	tool	Screening	ML		Automated								
3037	Santos23	2023	[84]	EPPI-Reviewer	tool	Identification, Screening	ML										
3037	Santos23	2023	[84]	ExaCT	tool	Data Extraction	ML: Text Mining		Automated								
3037	Santos23	2023	[84]	GAPScreener	tool	Screening	ML		Semi-automated						Human genetics		
3037	Santos23	2023	[84]	IBM Parse	tool	Identification	ML: Text Mining		Semi-automated								
3037	Santos23	2023	[84]	LitSuggest	tool	Identification	ML		Semi-automated								
3037	Santos23	2023	[84]	MetaPreg	tool	Identification, Screening, Data Extraction	ML: NLP		Semi-automated						Pregnancy meds		

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ID	Study	Year	Ref	Tool Name	Type	PRISMA Support	AI Type	Performance Metrics	Automation	Integration	Input Type	Source	Import/Export	Biblio	Domain	Output	Effectiveness
3037	Santos23	2023	[84]	MMiDas-AE	tool	Screening	NLP		Semi-automated								
3037	Santos23	2023	[84]	NaCTeM	tool	Data Extraction	ML: Text Mining		Automated						Genes		
3037	Santos23	2023	[84]	Rayyan	tool	Screening	ML		Semi-automated								
3037	Santos23	2023	[84]	ResearchScreener	tool	Screening	ML		Semi-automated								
3037	Santos23	2023	[84]	Revtools	tool	Data Synthesis	NLP: Text Mining		Automated								
3037	Santos23	2023	[84]	RobotAnalyst	tool	Screening	ML		Automated								
3037	Santos23	2023	[84]	RobotReviewer	tool	Data Extraction, Bias	ML		Semi-automated						RCTs		
3037	Santos23	2023	[84]	RobotSearch	tool	Identification	ML: NLP										
3037	Santos23	2023	[84]	Screen4Me	tool	Identification	ML								RCTs		
3037	Santos23	2023	[84]	SRToolBox	tool	Identification, Screening, Data Extraction, Data Synthesis	ML		Automated								
3037	Santos23	2023	[84]	SWFIT-Active Screeer	tool	Screening	ML		Automated								
3037	Santos23	2023	[84]	SWFIT-Review	tool	Identification	ML		Automated								
3037	Santos23	2023	[84]	Twister	tool	Identification	ML: Text Mining		Semi-automated								
3038	Alshami23	2023	[3]	ChatGPT3.5-based screening	tool	Identification, Screening, Data Extraction, Data Synthesis	LLM	Classification, F1-score	Automated	Standalone	Text	Closed	No	No		Text	91% classification, 88% F1
3060	Wang23	2023	[104]	ChatGPT-based screening	tool	Identification	LLM	Precision, recall, F1-score, F3-score	Semi-automated	Standalone	Text	Closed	No	No		Text	
3157	Scott23	2023	[86]	2weekSR	tool	Identification, Screening, Data Extraction	ML		Semi-automated	Integrated							
3342	Manning23	2023	[44]	Elicit	tool	Identification, Screening	ML		Semi-automated	Standalone	Text (prompt)	Closed				Text	
3342	Manning23	2023	[44]	ResearchRabbit	tool	Identification, Screening	NLP		Semi-automated	Standalone	Text (prompt)	Closed				Text	
3621	Sundaram23	2023	[91]	NLP/TM-based screening	tool	Data Extraction, Data Synthesis	NLP: Text Mining		Semi-automated	Standalone							
3679	Faria23	2023	[25]	Transformer	tool	Screening	ML	Workload saved (WSS@R)	Semi-automated	Standalone	Papers						5.8
3780	Hasny23	2023	[37]	srBERT	tool	Screening	ML: NLP	Recall, F1-score, screening reduction	Automated	Integrated	Abstract, titles	Open	No	No		Text	ee table
3814	Wilson23	2023	[106]	In-Vitro screening	tool	Screening	ML: Text Mining	Sensitivity, specificity, precision	Automated	Standalone	Abstract, titles, full text	Closed	No	No		Text	0.972, 0.980, 0.764
3993	O'Keefe23	2023	[69]	Anne O'Tate	tool	Identification	ML: Text Mining		Semi-automated	Integrated	Search string (text)					Tabular	
3993	O'Keefe23	2023	[69]	BiblioShiny	tool	Identification	ML: Text Mining		Semi-automated	Integrated	Scopus/WoS/PubMed file		Yes			Visualizations	
3993	O'Keefe23	2023	[69]	Carrot2	tool	Identification	ML: Text Mining		Semi-automated	Standalone	Search string (text)		Yes			Visualizations	
3993	O'Keefe23	2023	[69]	CitNetExplorer	tool	Identification	ML: Text Mining		Semi-automated	Standalone	WoS file		Yes			Network maps	
3993	O'Keefe23	2023	[69]	EndNote	tool	Identification	ML: Text Mining		Semi-automated	Standalone	Papers		Yes			Tabular	
3993	O'Keefe23	2023	[69]	Keyword-Analyzer	tool	Identification	ML: Text Mining		Semi-automated	Standalone	Text		Yes			Tabular	
3993	O'Keefe23	2023	[69]	Lingo3G (Carrot)	tool	Identification	ML: Text Mining		Semi-automated	Standalone	Search string (text)		Yes			Visualizations	
3993	O'Keefe23	2023	[69]	Lingo4G	tool	Identification	ML: Text Mining		Semi-automated	Standalone	Search string (text)		Yes			Visualizations	

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ID	Study	Year	Ref	Tool Name	Type	PRISMA Support	AI Type	Performance Metrics	Automation	Integration	Input Type	Source	Import/Export	Biblio	Domain	Output	Effectiveness
3993	O'Keefe23	2023	[69]	MeSHonDemand	tool	Identification	ML: Text Mining		Semi-automated	Standalone	Full text		Yes			Tabular	
3993	O'Keefe23	2023	[69]	PubReMiner	tool	Identification	ML: Text Mining		Semi-automated	Standalone	Search string (text)		Yes			Tabular	
3993	O'Keefe23	2023	[69]	TerMine	tool	Identification	ML: 3993	O'Keefe23	2023	[69]	Text Analyzer	tool	Identification	ML: Text Mining		Semi-automated	Standalone
Text		Yes			Tabular												
3993	O'Keefe23	2023	[69]	Tm for R	tool	Identification	ML: Text Mining		Semi-automated	Integrated	Papers (PDF)		Yes			Tabular	
3993	O'Keefe23	2023	[69]	VosViewer	tool	Identification	ML: Text Mining		Semi-automated	Standalone	DOI/JSON/Scopus		Yes			Network maps	
3993	O'Keefe23	2023	[69]	Voyant	tool	Identification	ML: Text Mining		Semi-automated	Standalone	Full text		Yes			Multiple visualizations	
3993	O'Keefe23	2023	[69]	Yale MeSH Analyser	tool	Identification	ML: Text Mining		Semi-automated	Standalone	PubMed IDs		Yes			Tabular	
3994	Ammirato22	2022	[5]	MySLR	tool	Identification, Screening, Data Extraction, Data Synthesis	ML: Text Mining, NLP		Semi-automated	Integrated							
4117	Adam22	2022	[2]	Pythia	tool	Screening	ML: NLP	Precision, sensitivity	Semi-automated	Standalone	Abstract, titles		No	No			sensitivity 57%
4174	Facchinetti22	2022	[22]	slr-kit	framework	Data Extraction, Bias, Analysis, QA	ML	Precision, recall	Semi-automated	Integrated	Abstract, title		Yes	Yes		Papers	see table
4387	Bernardes22	2022	[9]	ASReview	tool	Screening	ML		Semi-automated	Standalone	Abstract, title				Toxicology		
4413	Walker22	2022	[101]	Dextr	tool	Data Extraction	ML	Precision, recall	Semi-automated	Standalone	Full text	Closed	Yes	No	Public health	Text	96%, 91.8%
4730	Bulk22	2022	[98]	ML-based screening	tool	Screening	ML	Precision, recall, F1-score	Automated	Standalone	Abstract, title	Closed	No	No	Food consumption	Classification	57.8% irrelevant reduction
4746	Hannousse22	2022	[35]	Document Screener	tool	Screening	ML	F1-score, accuracy	Semi-automated	Standalone	Metadata	Open	No	No		Classification	89% F1, 97% accuracy
4782	Kusa22	2022	[50]	Citation Screener	tool	Screening	ML	Included, excluded, WSS	Automated	Standalone	Citations		No	No	Multiple		Dataset dependent
4842	Aum21	2021	[8]	srBert	tool	Screening	ML	Accuracy, F1-score, AUC	Automated	Standalone	Abstract, title	Closed	No	No		Papers	94.35% accuracy
4893	Holub21	2021	[38]	AutoLit	tool	Data Extraction	ML		Semi-automated				No	No			
4901	Chai21	2021	[15]	Research Screener	tool	Screening	ML	Workload saving	Semi-automated	Standalone	Abstract, title		Yes	No			60-96% across SLRs
5156	Noel-Storr21	2021	[65]	Screen4Me	tool	Screening	ML	Agreement, sensitivity, specificity	Semi-automated	Integrated	Abstract, title		Yes	No		Classification	100% sensitivity
5275	Haastrecht21	2021	[99]	SYMBALS	tool	Screening	ML: Active Learning	Speed	Automated	Standalone	Abstract, title	Open	No	No			6x faster
5525	Dhrangadhariya20	2020	[6]	Domain specific Citation Screening	tool	Screening	ML	Precision, recall, F1, AUC	Automated	Standalone	Abstract, titles		No	No	Multiple		Algorithm dependent
5565	Pradhan19	2019	[74]	EXACT	tool	Data Extraction	AI: Data Mining		Semi-automated	Standalone	Full text		Yes	Yes		Spreadsheet	
5579	Gates20	2020	[27]	Abstrackr	tool	Screening	ML	Precision, time saved	Semi-automated	Standalone	Abstract, titles		Yes	No	Multiple	Tabular	3-82 hours saved
5822	Marshall19	2019	[58]	RobotSearch	tool	Identification	ML		Automated								
5822	Marshall19	2019	[58]	Cochrane Register	tool	Identification	ML										
5822	Marshall19	2019	[58]	RCT Tagger	tool	Identification	ML										
5822	Marshall19	2019	[58]	Thalia	tool	Identification	ML										
5822	Marshall19	2019	[58]	Abstrackr	tool	Screening	ML										
5822	Marshall19	2019	[58]	EPPI Reviewer	tool	Screening	ML										
5822	Marshall19	2019	[58]	RobotAnalyst	tool	Screening	ML										
5822	Marshall19	2019	[58]	SWIFT-Review	tool	Screening	ML										
5822	Marshall19	2019	[58]	Colandr	tool	Screening	ML										
5822	Marshall19	2019	[58]	Rayyan	tool	Screening	ML										
5822	Marshall19	2019	[58]	ExACT	tool	Data Extraction	ML										
5822	Marshall19	2019	[58]	RobotReviewer	tool	Data Extraction	ML										

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5822	Marshall19	2019	[58]	NaCTeM	tool	Data Extraction	ML: Text Mining										
5822	Marshall19	2019	[58]	RobotReviewer	tool	Bias	ML										
5870	Schmidt19	2019	[85]	RAPTOR: RevMan	tool	Data Extraction	ML		Automated	Integrated	Full text	Open	Yes	No		XML	
5897	Lerner19	2019	[53]	Automatic NMA Screener	tool	Screening	ML: NLP	Sensitivity, workload decrease	Automated	Standalone	Citations	Open code	No	No	Medical	Classification	100% sensitivity
5898	Martin19	2019	[59]	Trial2rev	tool	Identification, Screening, Data Extraction	ML		Semi-automated	Standalone		Open	Yes	Yes			
5906	Hu18	2018	[40]	MetaCyto	tool	Data Extraction	ML		Automated	Standalone	Full text		No	No	Cytometry		
5916	Waffenschmidt18	2018	[100]	Abstrackr	tool	Screening	ML		Semi-automated	Standalone							
5916	Waffenschmidt18	2018	[100]	Rayyan	tool	Screening	ML		Semi-automated	Standalone							
5916	Waffenschmidt18	2018	[100]	EPPI Reviewer	tool	Screening	ML		Semi-automated	Integrated							
5957	Zdraveski19	2019	[111]	NLP Toolkit	tool	Identification, Screening	ML: NLP		Automated	Integrated	Abstract, titles	Open	Yes	No	Assisted living	Multiple	
5964	Goswami19	2019	[30]	Data Extracter	tool	Data Extraction	ML	Recall, precision	Automated	Standalone	Abstract, titles		No	No	Anxiety measures	Sentences	100% recall
6006	Kreiner18	2018	[48]	Twister	tool	Screening	ML: Text Mining		Semi-automated	Standalone	Abstract, titles	Open	Yes	No	Clinical guidelines	PDF	
6090	Gates18	2018	[28]	Abstrackr	tool	Screening	ML	Sensitivity, specificity, precision	Semi-automated	Standalone	Abstract, titles	Closed	Yes	No	Health		75% sensitivity
6098	Tsafnat18	2018	[97]	KeyCharacteristic Screener	tool	Screening	ML: Text Mining	Recall, exclusion	Automated	Standalone	Abstract, titles	Closed	No	No			23.7% excluded
6132	Howard16	2016	[39]	SWIFT-Review	tool	Screening	ML	WSS@R	Semi-automated	Standalone	Abstract, titles		Yes	Yes			Dataset dependent
6331	Bowes12	2012	[12]	SLuRp	tool												
6369	Garcia14	2014	[26]	Text-Classifier	tool	Screening	ML	Precision, recall	Semi-automated	Standalone	Abstract, titles				Medicine	Classification	Up to 84%
6421	Wallace10	2010	[102]	Citation Screener	tool	Screening	ML	Screening reduction	Semi-automated	Standalone	Abstract, titles				Biomedical	Citations	70% reduction